

## Questions

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## Q1 - 2024 (04 Apr Shift 1)

Which of the following nuclear fragments corresponding to nuclear fission between neutron ( ${}^1_0\text{n}$ ) and uranium isotope ( ${}^{235}_{92}\text{U}$ ) is correct :

(1)  ${}^{144}_{56}\text{Ba} + {}^{89}_{36}\text{Kr} + 4{}^1_0\text{n}$

(2)  ${}^{144}_{56}\text{Ba} + {}^{89}_{36}\text{Kr} + 3{}^1_0\text{n}$

(3)  ${}^{140}_{56}\text{Xe} + {}^{94}_{38}\text{Sr} + 3{}^1_0\text{n}$

(4)  ${}^{153}_{51}\text{Sb} + {}^{99}_{41}\text{Nb} + 3{}^1_0\text{n}$

## Q2 - 2024 (04 Apr Shift 2)

The disintegration energy  $Q$  for the nuclear fission of  ${}^{235}\text{U} \rightarrow {}^{140}\text{Ce} + {}^{94}\text{Zr} + n$  is \_\_\_\_\_ MeV.

Given atomic masses of  ${}^{235}\text{U} : 235.0439u$ ;  ${}^{140}\text{Ce} : 139.9054u$ ,

${}^{94}\text{Zr} : 93.9063u$ ;  $n : 1.0086u$ ,

Value of  $c^2 = 931\text{MeV/u}$

## Q3 - 2024 (05 Apr Shift 1)

If three helium nuclei combine to form a carbon nucleus then the energy released in this reaction is \_\_\_\_\_

$\times 10^{-2}\text{MeV}$ . (Given  $1u = 931\text{MeV}/c^2$ , atomic mass of helium =  $4.002603u$ )

## Q4 - 2024 (08 Apr Shift 1)

Binding energy of a certain nucleus is  $18 \times 10^8 \text{ J}$ . How much is the difference between total mass of all the nucleons and nuclear mass of the given nucleus:

(1)  $10\mu\text{g}$

(2)  $20\mu\text{g}$

(3)  $0.2\mu\text{g}$

(4)  $2\mu\text{g}$

## Q5 - 2024 (08 Apr Shift 2)

## Questions

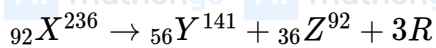
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If  $M_o$  is the mass of isotope  ${}^{12}_5B$ ,  $M_p$  and  $M_n$  are the masses of proton and neutron, then nuclear binding energy of isotope is :

- (1)  $(M_o - 5M_p) C^2$
- (2)  $(5M_p + 7M_n - M_o) C^2$
- (3)  $(M_o + 12M_n) C^2$
- (4)  $(M_o - 5M_p - 7M_n) C^2$

## Q6 - 2024 (08 Apr Shift 2)

In a hypothetical fission reaction



The identity of emitted particles ( R ) is :

- (1) Electron
- (2) Neutron
- (3)  $\gamma$ -radiations
- (4) Proton

## Q7 - 2024 (09 Apr Shift 1)

The energy equivalent of 1 g of substance is :

- (1)  $5.6 \times 10^{12} \text{ MeV}$
- (2)  $5.6 \times 10^{26} \text{ MeV}$
- (3)  $11.2 \times 10^{24} \text{ MeV}$
- (4)  $5.6 \text{ eV}$

## Q8 - 2024 (09 Apr Shift 1)

A star has 100% helium composition. It starts to convert three  ${}^4\text{He}$  into one  ${}^{12}\text{C}$  via triple alpha process as  ${}^4\text{He} + {}^4\text{He} + {}^4\text{He} \rightarrow {}^{12}\text{C} + Q$ . The mass of the star is  $2.0 \times 10^{32} \text{ kg}$  and it generates energy at the rate of

## Questions

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$5.808 \times 10^{30}$  W. The rate of converting these  ${}^4\text{He}$  to  ${}^{12}\text{C}$  is  $n \times 10^{42} \text{ s}^{-1}$ , where  $n$  is \_\_\_\_\_

[ Take, mass of  ${}^4\text{He} = 4.0026\text{u}$ , mass of  ${}^{12}\text{C} = 12\text{u}$  ]

## Q9 - 2024 (09 Apr Shift 2)

A nucleus at rest disintegrates into two smaller nuclei with their masses in the ratio of 2 : 1. After disintegration they will move :

- (1) in the same direction with same speed.
- (2) in opposite directions with the same speed.
- (3) in opposite directions with speed in the ratio of 2 : 1 respectively.
- (4) in opposite directions with speed in the ratio of 1 : 2 respectively.

## Q10 - 2024 (09 Apr Shift 2)

The energy released in the fusion of 2 kg of hydrogen deep in the sun is  $E_H$  and the energy released in the fission of 2 kg of  ${}^{235}\text{U}$  is  $E_U$ . The ratio  $\frac{E_H}{E_U}$  is approximately:

(Consider the fusion reaction as  $4 {}^1_1\text{H} + 2e^- \rightarrow {}^4_2\text{He} + 2\nu + 6\gamma + 26.7\text{MeV}$ , energy released in the fission reaction of  ${}^{235}\text{U}$  is  $200\text{MeV}$  per fission nucleus and  $N_A = 6.023 \times 10^{23}$ )

- (1) 7.62
- (2) 25.6
- (3) 15.04
- (4) 9.13

Questions

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Answer Key

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Q1 (2) mathongo // m Q2 (208) // mathongo Q3 (727) mathongo // ma Q4 (2) // mathongo

Q5 (2) athongo // m Q6 (2) go // mathongo Q7 (2) mathongo // ma Q8 (15) // mathongo

Q9 (4) athongo // m Q10 (1) // mathongo // mathongo // mathongo // mathongo

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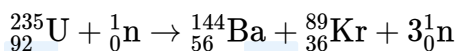
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## Solutions

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Q1

Balancing mass number and atomic number



Q2



Disintegration energy

$$Q = (m_R - m_p) c^2$$

$$m_R = 235.0439\text{u}$$

$$m_p = 139.9054\text{u} + 93.9063\text{u} + 1.0086\text{u}$$

$$= 234.8203\text{u}$$

$$Q = (235.0439\text{u} - 234.8203\text{u})c^2$$

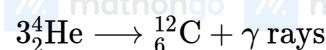
$$= 0.2236c^2$$

$$= 0.2236 \times 931$$

$$Q = 208.1716$$

Q3

Reaction :



$$\text{Mass defect} = \Delta m = (3 m_{\text{He}} - m_{\text{C}})$$

$$= (3 \times 4.002603 - 12) = 0.007809\text{u}$$

Energy released

$$= 931\Delta m \text{MeV}$$

$$= 7.27\text{MeV} = 727 \times 10^{-2}\text{MeV}$$

Q4

$$\Delta mc^2 = 18 \times 10^8$$

$$\Delta m \times 9 \times 10^{16} = 18 \times 10^8$$

$$\Delta m = 2 \times 10^{-8} \text{ kg} = 20\mu\text{g}$$

Q5

$$\text{B.E.} = \Delta m C^2$$

$$(5M_p + 7M_n - M_o) C^2$$

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## Solutions

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Q6

$$Z \text{ in LHS} = 92$$

$$Z \text{ in RHS} = 56 + 36 = 92$$

$$A \text{ in LHS} = 236$$

$$A \text{ in RHS} = 141 + 92 = 233$$

So 3 neutrons are released.

Q7

$$E = mC^2$$

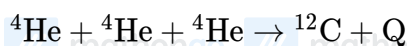
$$\Rightarrow E = (1 \times 10^{-3}) \times (3 \times 10^8)^2 \text{ J}$$

$$\Rightarrow E = (10^{-3}) (9 \times 10^{16}) (6.241 \times 10^{18}) \text{ eV}$$

$$E = 56.169 \times 10^{31} \text{ eV}$$

$$E \approx 5.6 \times 10^{26} \text{ MeV}$$

Q8



$$\text{power generated} = \frac{N}{t} Q$$

where,  $N \rightarrow$  No. of reaction /sec.

$$Q = (3 m_{\text{He}} - m_{\text{C}}) C^2$$

$$Q = (3 \times 4.0026 - 12)(3 \times 10^8)^2$$

$$Q = 7.266 \text{ MeV}$$

$$\frac{N}{t} = \frac{\text{power}}{Q} = \frac{5.808 \times 10^{30}}{7.266 \times 10^6 \times 1.6 \times 10^{-19}}$$

$$\frac{N}{t} = 5 \times 10^{42}$$

rate of conversion of  ${}^4\text{He}$  into  ${}^{12}\text{C} = 15 \times 10^{42}$  Hence,  $n = 15$

Q9

By conservation of momentum

$$P_i = P_f$$

$$0 = m_1 u_1 + m_2 u_2$$

$$\frac{u_1}{u_2} = - \left[ \frac{1}{2} \right] \text{ as } \frac{m_1}{m_2} = \frac{2}{1}$$

move in opposite direction with speed ratio 1 : 2

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## Solutions

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Q10

In each fusion reaction,  $4 {}_1^1\text{H}$  nucleus are used.

Energy released per Nuclei of  ${}_1^1\text{H} = \frac{26.7}{4} \text{MeV}$

$\therefore$  Energy released by 2 kg hydrogen ( $E_H$ )

$$= \frac{2000}{1} \times N_A \times \frac{26.7}{4} \text{MeV}$$

&amp;

$\therefore$  Energy released by 2 kg Vranium ( $E_V$ )

$$= \frac{2000}{235} \times N_A \times 200 \text{MeV}$$

So,

$$\frac{E_H}{E_V} = 235 \times \frac{26.7}{4 \times 200} = 7.84$$

$\therefore$  Approximately close to 7.62

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